Evaluating calibration-free two-view 3D pose fusion for exploratory repeated-cycle gymnastics analysis

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Abstract

Markerless three-dimensional motion analysis from ordinary video is attractive for sports settings, but monocular estimates remain dependent on camera view, occlusion and depth ambiguity. We evaluated how reliably two temporally aligned monocular pose streams can be reused when camera calibration and independent three-dimensional labels are unavailable. Face- and side-view poses were transformed to a pelvis-centred body frame, after which a view-order-invariant temporal network predicted a bounded correction using rotation, skeletal and temporal self-supervision. In 137 participants performing a repeated gymnastics movement, body-frame averaging reduced mean distance to a same-video triangulated pseudo-reference by 47.54% relative to world-coordinate averaging; several aligned deterministic variants performed similarly. On 14 held-out participants, the complete learned model reached 60.78 mm mean per-joint position error after sequence-level similarity alignment and was statistically indistinguishable from arithmetic averaging (60.85 mm) and the preceding learned ablation (60.80 mm). External testing on 199 Unity samples showed no zero-shot advantage over direct three-dimensional averaging (178.506 versus 166.537 mm), whereas calibrated two-dimensional triangulation reached 30.259 mm. Person-disjoint inference yielded 928 cycles from 80 elderly-labeled and 57 student-labeled participants. Cohort contrasts in angular speed and dimensionless jerk changed materially with the pose source. Coordinate canonicalization, rather than the learned correction, provided the clearest in-domain engineering gain. The resulting representation supports exploratory sequence analysis, not absolute biomechanical measurement or causal conclusions about ageing.

Keywords: markerless motion capture, self-supervised learning, data fusion, human movement, trunk rotation, repeated measures

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1 Introduction

Three-dimensional (3D) motion analysis can support technique assessment, screening and feedback in sport, but laboratory motion capture is not always available where movements are practised. Video-based systems lower the capture burden. Modern monocular estimators can recover a dense articulated body from one ordinary camera; SAM 3D Body, for example, predicts a full-body mesh and Momentum Human Rig skeleton from an image [1]. A monocular 3D estimate nevertheless inherits depth ambiguity, self-occlusion, motion blur, truncation and uncertain scale. Face and side cameras observe different failures, so two predicted pose streams are potentially complementary but cannot be averaged safely in their original coordinates.

Most multi-view methods combine images or two-dimensional (2D) evidence using known or estimated camera geometry [2–5]. Other self-supervised approaches learn from multi-view consistency but still begin from images or 2D poses [6–8]. These methods do not directly address a common applied situation: complete monocular 3D sequences have already been generated, temporal correspondence is known, but calibration and independent 3D labels are unavailable. Reprocessing the images with a calibrated system may be impossible after field collection.

The engineering problem is especially relevant to rotation-dominant movement. An output can be visually smooth and have low coordinate disagreement while attenuating the trunk rotation or peak angular speed of interest. Evaluation must therefore distinguish coordinate harmonization from learned correction and determine whether downstream movement findings remain stable when the pose representation changes.

This study evaluates three linked questions. First, how much of the in-domain agreement with a same-video pseudo-reference follows from coordinate canonicalization rather than learned fusion? Second, does the complete self-supervised model satisfy its declared symmetry, residual and cycle-motion constraints while maintaining held-out agreement, and how far does it transfer to independent native 3D? Third, are between-cohort and within-person repeated-cycle findings stable across pose sources? We contribute: (1) an applied post-estimation fusion formulation; (2) a pelvis-centred, rotation-aware and view-order-invariant temporal model trained with self-supervision; (3) a leakage-controlled evidence hierarchy separating same-video pseudo-reference agreement from limited external native-3D testing; and (4) a downstream representation-sensitivity case study using mixed models. Because independent motion capture and participant-level demographic covariates are unavailable, the study does not claim absolute biomechanical accuracy, clinical validity or causal effects of ageing.

2 Methods

2.1 Study data and processing workflow

The private dataset comprised paired face- and side-view videos from 137 participants performing repeated cycles of one gymnastics movement. Dataset labels identified 80 participants as elderly and 57 as students. Exact ages, sex, health status, anthropometry and training history were not available for the present analysis; the terms *elderly-labeled* and *student-labeled* are therefore used as observational group names.

Each video was processed independently with the fixed SAM 3D Body estimator to produce $J = 70$ 3D keypoints and validity masks. An upstream procedure estimated one side-to-face temporal offset for each participant and recorded the frame bounds of each complete cycle. Fusion used this stored offset without re-estimating correspondence. A separate calibrated pipeline triangulated SAM 3D Body 2D keypoints from the same videos. Triangulated sequences were accessible only to the evaluation software and were never read during fusion training, validation, checkpoint selection or inference. They are consequently treated as a same-video *pseudo-reference*, not independent ground truth.

Ethics statement—author verification required before submission. The institutional ethics approval or exemption identifier and the confirmed participant-consent statement were not recoverable from the analysis repository. The author must replace this sentence with the approved wording before the manuscript is submitted.

2.2 Fusion method

Let $\mathbf{X}^f, \mathbf{X}^s \in \mathbb{R}^{T\times J\times 3}$ denote temporally aligned face- and side-view keypoints for a trial. Each view is canonicalized independently. The pelvis origin $\mathbf{o}_t^v$ is removed, and orthonormal axes are constructed from the hip direction and the pelvis-to-thorax direction. The robust trial scale a^v is the median valid pelvis–thorax distance. With body-frame rotation $\mathbf{R}_{p,t}^v \in \mathrm{SO}(3)$, the canonical point is

$$\widetilde{\mathbf{x}}_{t,j}^v = \frac{(\mathbf{x}_{t,j}^v - \mathbf{o}_t^v)\mathbf{R}_{p,t}^v}{a^v}. \tag{1}$$

The row-vector convention in Eq. 1 maps each camera-dependent pose into a pelvis-centred, trial-scaled body frame without estimating camera geometry. Degenerate transforms retain the last observable frame for continuity but remain masked from supervision.

A thorax frame is constructed from the shoulder direction and a thorax-to-neck direction. Relative pelvis–thorax rotation is $\mathbf{R}_{ph,t}^v = (\mathbf{R}_{p,t}^v)^\top \mathbf{R}_{h,t}^v$. Signed axial rotation and circular finite differences provide angle, angular velocity and angular acceleration. Physical timestamps, rather than an assumed constant frame interval, are used for derivatives.

Fixed frame quality combines valid-joint coverage with robust median-absolute-deviation penalties for shoulder width, hip width, torso length, bone rigidity, angular acceleration and a degenerate trunk frame. For quality q_t^v and joint-validity mask $M_{t,j}^v$,

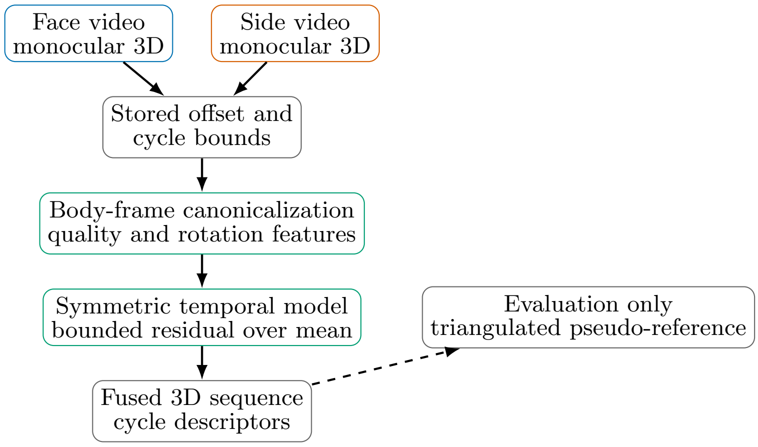

Fig. 1 Applied workflow and supervision boundary. Triangulated 3D is resolved only by the evaluation layer and cannot influence model training or selection.

the conservative base pose is

$$\mathbf{B}_{t,j} = \frac{q_t^f M_{t,j}^f \widetilde{\mathbf{x}}_{t,j}^f + q_t^s M_{t,j}^s \widetilde{\mathbf{x}}_{t,j}^s}{q_t^f M_{t,j}^f + q_t^s M_{t,j}^s + \epsilon}. \tag{2}$$

A point observed in only one view passes through unchanged. Equal weighting is used if both points are valid but both quality values are zero.

Shared encoders process per-view canonical coordinates, velocities, masks, quality and pelvis–thorax rotation features. View-order invariance is imposed by retaining only the encoded mean and absolute difference, together with symmetric cross-view disagreement. Six non-causal residual temporal convolutional blocks use dilations 1, 2, 4, 8, 16 and 32, following the long-context principle of temporal convolutional pose models [9, 10]. The network predicts a per-joint correction bounded to 0.05 normalized trial units:

$$\widehat{\mathbf{x}}_{t,j} = \mathbf{B}_{t,j} + 0.05 \tanh(\mathbf{h}_{t,j}). \tag{3}$$

The residual is disabled unless both views are valid. The fused canonical pose is finally mapped back using the face-view origin, orientation and scale; this is an uncalibrated face reference, not a physical world coordinate system. Here, *auditable* denotes four inspectable properties: view-order invariance, a bounded residual, explicit missing-view pass-through and reported departure in cycle range and peak speed from the deterministic base. It does not imply biomechanical validity.

2.3 Self-supervision and training

Clean cross-view observations form pseudo-targets only where evidence is usable: closely agreeing points are quality-weighted, one clearly higher-quality view is selected, and single-view observations are retained when the other view is missing. Large unexplained disagreements are excluded. Seven reproducible corruptions—joint dropout, block dropout, impulse noise, random-walk drift, upper-body rotation bias, frozen segments and integer frame shifts—are applied without overwriting the clean streams.

The complete model minimizes nine masked losses: corruption recovery, high-consensus identity, axial-angle agreement, full relative-rotation agreement, bone-length consistency, temporal bone rigidity, adaptive acceleration agreement, minimal residual magnitude and complete-cycle axial range. The complete-cycle term is evaluated on unpadded cycles, whereas other terms use 128-frame windows. The model has 128 hidden channels and was optimized with Adam for 100 epochs at learning rate 10^{-3} and batch size 32. Checkpoint selection averaged fixed-corruption recovery, bone variation, rotation consistency, identity and cycle-range scores; none used the triangulated pseudo-reference.

2.4 Evaluation hierarchy

The fixed person split contained 96 training, 27 validation and 14 held-out test participants. The 14-person set was the primary learned comparison. Person-level mean per-joint position error was calculated after one static sequence-level similarity transform and hip centring against the triangulated pseudo-reference. Each comparison was paired against the complete model on the same participants using a person-bootstrap 95% confidence interval and a two-sided Wilcoxon signed-rank test; Holm correction covered the nine planned comparisons. Results use one training seed and therefore do not quantify optimization variance.

A deterministic analysis over all 137 participants first isolated the effect of coordinate representation by comparing world-coordinate, root-aligned, similarity-aligned and body-frame fusion variants. The held-out learned comparison then tested whether residual learning added evidence beyond the canonical deterministic means. One legacy row whose joint weights were fitted to the evaluation pseudo-reference was retained only as a leakage diagnostic and was excluded from recommendations.

Limited external testing used 199 rendered samples from three sequences, one Unity avatar/environment and a fixed two-camera configuration. Sixteen mapped joints were evaluated against native Unity 3D after sequence-level similarity alignment. The private model was applied zero-shot. Calibrated triangulation used 2D observations and exact Unity cameras, so it is reported as a different input regime rather than a directly matched fusion baseline.

2.5 Downstream representation-sensitivity case study

Ten cohort-stratified outer folds separated people, not cycles, so every participant received an out-of-fold prediction. The resulting 928 eligible cycles comprised 539 elderly-labeled and 389 student-labeled cycles. Each cycle was direction-aligned and resampled to 101 phase points. Eight prespecified descriptors covered axial rotation range, 95th-percentile angular speed, peak-rotation phase, 95th-percentile trunk tilt, 95th-percentile wrist wrapping, cycle duration, log dimensionless angular jerk and leave-one-cycle-out whole-body repeatability.

For each descriptor, a cycle-level mixed model included cohort, normalized cycle position, their interaction and outer-fold fixed effects. Cycle position was centred

Table 1 Primary comparison on 14 held-out participants. Error is person-level mean ± standard deviation after one sequence-level similarity alignment to the same-video triangulated pseudo-reference. Differences are method minus the complete model (A6); p values are Holm-adjusted paired Wilcoxon tests.

ID	Method	Error (mm)	Difference [95% CI]	p_{Holm}
A0	Face only	77.92 ± 12.52	+17.14 [13.69, 21.34]	0.0011
A1	Side only	75.84 ± 13.93	+15.07 [10.23, 19.48]	0.0018
A2	Arithmetic mean	60.85 ± 8.98	+0.07 [−0.50, 0.59]	1.0000
A3	Quality mean	61.03 ± 8.84	+0.26 [0.17, 0.35]	0.0011
A4	Spatial objectives	62.81 ± 8.86	+2.04 [1.71, 2.36]	0.0011
A5	Rotation/temporal	60.80 ± 8.96	+0.02 [−0.38, 0.43]	1.0000
A6	Complete model	$\mathbf{60.78 \pm 8.72}$	—	—

at 0.5; therefore, the cohort coefficient estimates the elderly-minus-student contrast at the mid-repetition reference. Participant random intercepts and, when nonsingular, random cycle-position slopes accounted for repeated cycles. Holm correction was applied separately to the eight cohort coefficients and eight interactions. Within-person dispersion was the participant-level median absolute deviation (MAD); groups were compared with 10,000 participant-label permutations, participant-bootstrap confidence intervals and a separate eight-outcome Holm family. Pose-source sensitivity repeated the centred model using face-only, side-only and deterministic body-frame features.

3 Results

3.1 Coordinate-system effect and held-out learned comparison

Across all 137 participants, deterministic body-frame averaging had mean pseudo-reference error 0.0640 repository units, compared with 0.1221 for world-coordinate averaging, a 47.54% reduction against this deliberately uncorrected baseline. More informative leakage-free aligned variants occupied a narrow range from 0.0640 to 0.0663 with overlapping confidence intervals. The evidence therefore supports coordinate normalization, rather than a particular weighting rule or learned residual, as the main source of in-domain improvement.

Table 1 shows that A6 was statistically indistinguishable from arithmetic averaging and A5. Its complete-cycle range and peak-angular-speed retention relative to the deterministic quality mean were both 1.000, but these ratios assess departure from the base rather than physical validity. All canonical averaging and learned outputs were view-order invariant to the reported precision. Validation-only corruption diagnostics did not establish a general robustness advantage for A6: its output moved more than the deterministic quality mean under joint and temporal-block dropout. Freer post hoc residual variants increased error to 92.02–94.11 mm and are reported in the Online Resource. A6 is consequently retained as the prespecified complete test of the declared constraints, not as a demonstrated accuracy or robustness winner.

Table 2 Limited external evaluation against Unity native 3D. Direct-3D and calibrated-2D rows use different input information.

Input	Method	Error (mm)	Angle MAE (°)
Uncalibrated 3D	World average	166.537	47.497
Uncalibrated 3D	A6, zero-shot	178.506	45.983
Calibrated 2D	Triangulation	30.259	15.198

Table 3 Sensitivity of significant A6 cohort coefficients to pose source. Values are elderly-minus-student coefficients from the same centred mixed model; p values are Holm-adjusted within each source.

Outcome	OOF A6	Face	Side	Body-frame mean
Log angular speed	0.3006	−0.0111	0.0843	0.0891
p_{Holm}	0.0114	1.0000	0.0337	0.0323
Log dimensionless jerk	0.4233	0.0452	0.0646	0.0657
p_{Holm}	0.0001	1.0000	0.2475	0.2324

3.2 External native-3D test

The external result changed the interpretation (Table 2). A6 did not improve zero-shot positional transfer relative to direct 3D averaging. Calibrated 2D triangulation was substantially more accurate, but used camera geometry and evidence unavailable to the post-estimation method.

3.3 Representation sensitivity in the repeated-cycle case study

All eight mixed models converged. At mid-repetition, the elderly-labeled cohort had a 0.3006 higher log angular-speed coefficient (95% confidence interval 0.1136–0.4876, $p_{\mathrm{Holm}} = 0.0114$), corresponding to a ratio of 1.351, and a 0.4233 higher log dimensionless-jerk coefficient (0.2336–0.6130, $p_{\mathrm{Holm}} = 0.000098$). The other six corrected cohort contrasts were not significant. No cohort-by-cycle-position interaction survived correction; there was therefore no evidence of different monotonic first-to-last repetition trends. Within-person jerk MAD was 0.0377 higher in the elderly-labeled cohort (bootstrap 95% confidence interval 0.0135–0.0735, $p_{\mathrm{Holm}} = 0.0208$); no other within-person dispersion contrast survived.

Changing the pose source materially reduced the cohort coefficients (Table 3), especially for jerk. Separate participant-median permutation tests were also non-significant after correction for A6 angular speed and jerk. The stable finding from this case study is therefore the sensitivity of downstream inference to pose representation, not a representation-independent difference between the two labeled cohorts.

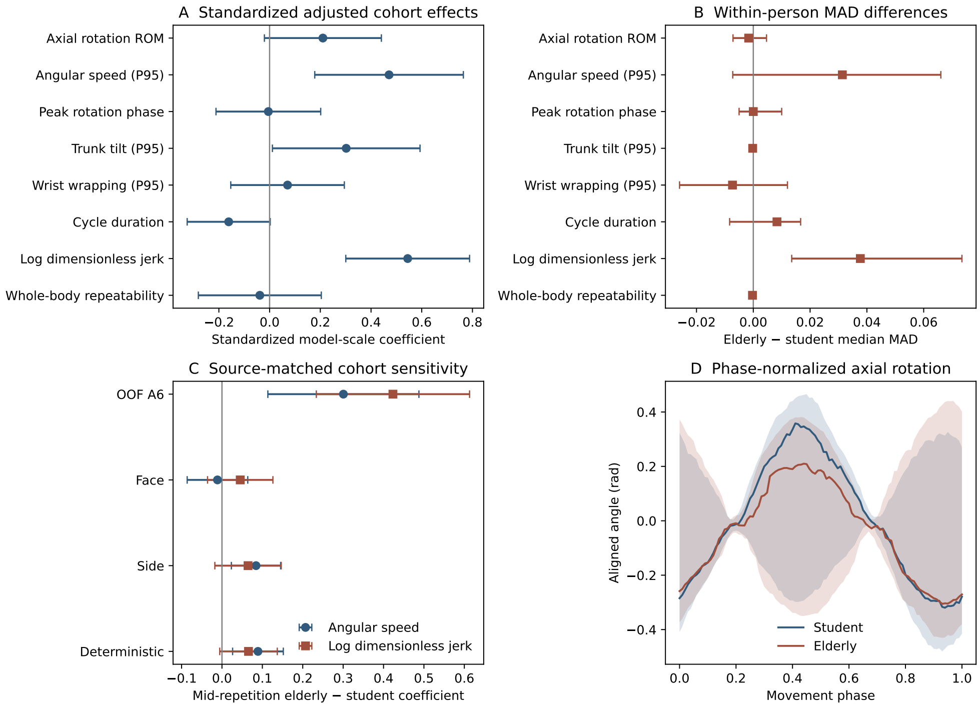

Fig. 2 Person-disjoint repeated-cycle analysis. (a) Standardized mid-repetition cohort effects. (b) Differences in participant-level within-person MAD. (c) Pose-source sensitivity for angular speed and dimensionless jerk. Error bars are 95% confidence intervals. (d) Participant-median phase-normalized axial rotation with interquartile bands.

4 Discussion

The most robust engineering result is the effect of the coordinate system. Mapping both monocular streams into a pelvis-centred body frame nearly halved their disagreement with the same-video pseudo-reference compared with world-coordinate averaging. Once this mismatch was removed, deterministic and learned methods occupied a narrow range. The held-out comparison provides no evidence that the complete network improves positional accuracy over arithmetic averaging or A5. This is evidence for coordinate harmonization, not learned fusion superiority. A6 instead serves as the prespecified complete test of explicit requirements: view-order invariance, bounded correction, missing-view fallback and complete-cycle rotation constraints.

This distinction matters when an applied system is selected. If the goal is a simple fused sequence and compute or training data are limited, deterministic body-frame averaging is the better-supported default. The learned model is justified when its explicit audit trail and constrained motion representation are required, but the current results do not justify extra complexity on position error or corruption stability. Reference-relative range and angular-speed retention also cannot establish biomechanical validity, because their denominator is the deterministic base rather than independent motion capture.

Unity exposes a second boundary. A6 transferred worse than direct 3D averaging for position, whereas calibrated triangulation from 2D evidence was far better. Projection geometry retains information that cannot reliably be recovered after two independent monocular 3D predictions have been made. In a prospective capture where camera calibration is feasible, calibrated reconstruction should therefore be preferred. Post-estimation fusion addresses archival or low-infrastructure data where calibration is unavailable; it is not a replacement for geometry.

The repeated-cycle experiment primarily demonstrates the risk of representation-dependent downstream inference. Person-disjoint prediction avoids evaluating a participant with a model trained on that participant, and mixed models retain cycle-level information while accounting for repeated observations. A6 produced cohort contrasts in angular speed and jerk and greater within-person jerk dispersion, but no group-specific repetition trend. The same models applied to other pose streams produced much smaller effects, and participant-median tests changed the inference. The signals may combine true movement differences, cohort composition, monocular estimation error and learned representation effects. They must not be interpreted as ageing effects, normative values, impairment markers or clinical findings.

The private pseudo-reference shares upstream 2D evidence with the candidate poses and is scored after sequence-level similarity alignment. Its errors can therefore be correlated with candidate errors, and the reported value is agreement with a construction rather than absolute accuracy. The primary learned test includes only 14 participants and one training seed. The Unity test contains one avatar/environment and fixed cameras. The cohort data lack individual demographic, health and training covariates. Finally, a single global time offset cannot model drift, dropped frames or rolling shutter, and the study covers one movement family and one face/side arrangement.

Future work should validate the representation against synchronized independent capture and across diverse people, cameras and movements. Public datasets such as Human3.6M, MPI-INF-3DHP and TotalCapture provide complementary settings [11–13]. For sports deployment, comparisons should include calibrated systems such as OpenCap, whose camera calibration, musculoskeletal modelling and laboratory validation support a different level of interpretation [14]. Repeated training seeds, local temporal alignment and measured participant covariates are also required before the cohort analysis can progress beyond hypothesis generation.

5 Conclusions

Calibration-free post-estimation fusion can provide a transparent sequence representation when only aligned monocular 3D pose streams remain available. Pelvis-centred canonicalization provided the clearest in-domain improvement; the complete self-supervised model matched but did not surpass strong held-out baselines, showed no general corruption-stability advantage and did not improve zero-shot external position accuracy. The repeated-cycle case study further showed that between-cohort and within-person inferences changed with the pose source. The workflow is therefore best treated as an exploratory, low-infrastructure representation layer. Absolute biomechanical, causal or clinical interpretation requires calibrated or independent reference capture and richer participant information.

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Data availability

The raw videos contain identifiable participant information and are not publicly available. Code and derived anonymized keypoint data have not yet been deposited in a public repository. Release or controlled-access conditions will be determined after institutional ethics, participant-consent and privacy review. Numerical results are retained in archived experiment artifacts.

Ethics approval and consent to participate

Author verification required before submission: insert the confirmed institutional ethics approval or exemption identifier and the participant consent statement. No ethics status is inferred from the software repository.

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